Measure sound speed

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Analyze sound travel measurements and their limitations.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 7.*
+4 mm; partly cancel.

02. Worked example: explain the model
Time covers both directions; range=v Δt/2.

03. Read and interpret the model
One-way; 340 m/s.

04. Apply the physics
v = 12/5 = 2.4 m/s. A larger time with the same distance gives a lower calculated speed.

05. Explain your reasoning
Relevant table; device latency/distance calibration/echo path ambiguity accepted.

06. Review: connect an earlier idea *Review: Unit 5, Day 10.*
The larger sphere has 1/4 of the smaller sphere’s area per mass.

07. Exit check
It uses half the actual sound path; answer half true value.

08. STRETCH: test the model
34 and 102 m; separation 68 m.

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01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 7.*
Resultant -9 mm. Energy also depends on motion, spatial gradients and the surrounding wave field.

02. Worked example: explain the model
Time covers both directions; range=v Δt/2.

03. Read and interpret the model
One-way; 340 m/s.

04. Apply the physics
340 m/s; lower 6.8/0.0205=331.7, upper 6.8/0.0195=348.7 m/s.

05. Explain your reasoning
Relevant table; device latency/distance calibration/echo path ambiguity accepted.

06. Review: connect an earlier idea *Review: Unit 5, Day 10.*
Area factor 16, mass factor 64; A/m becomes 1/4. Survival also depends on transfer rates, fragmentation and conditions, not only energy per mass.

07. Exit check
340 m/s; incorrect 242.9 m/s, biased low.

08. STRETCH: test the model
Central 340.9 m/s; bounds 1.5/0.0046≈326.1 to 1.5/0.0042≈357.1 m/s.